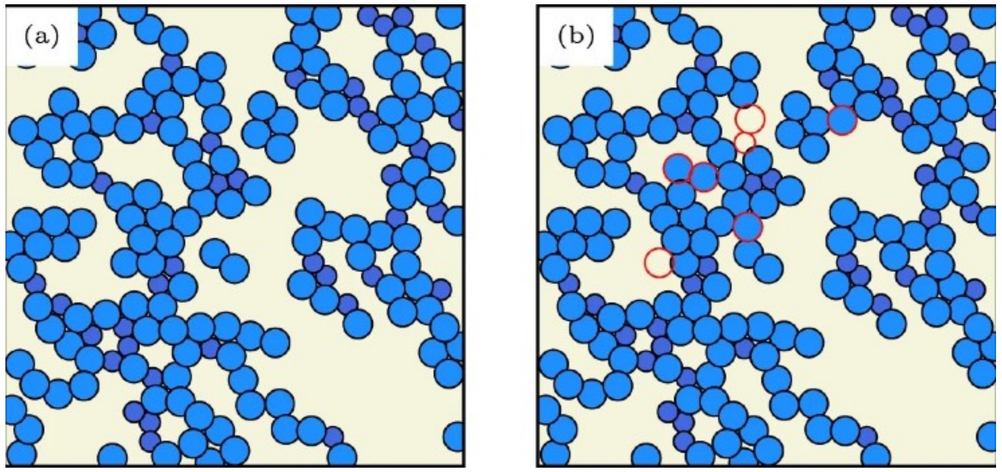
ML successes in disordered systems

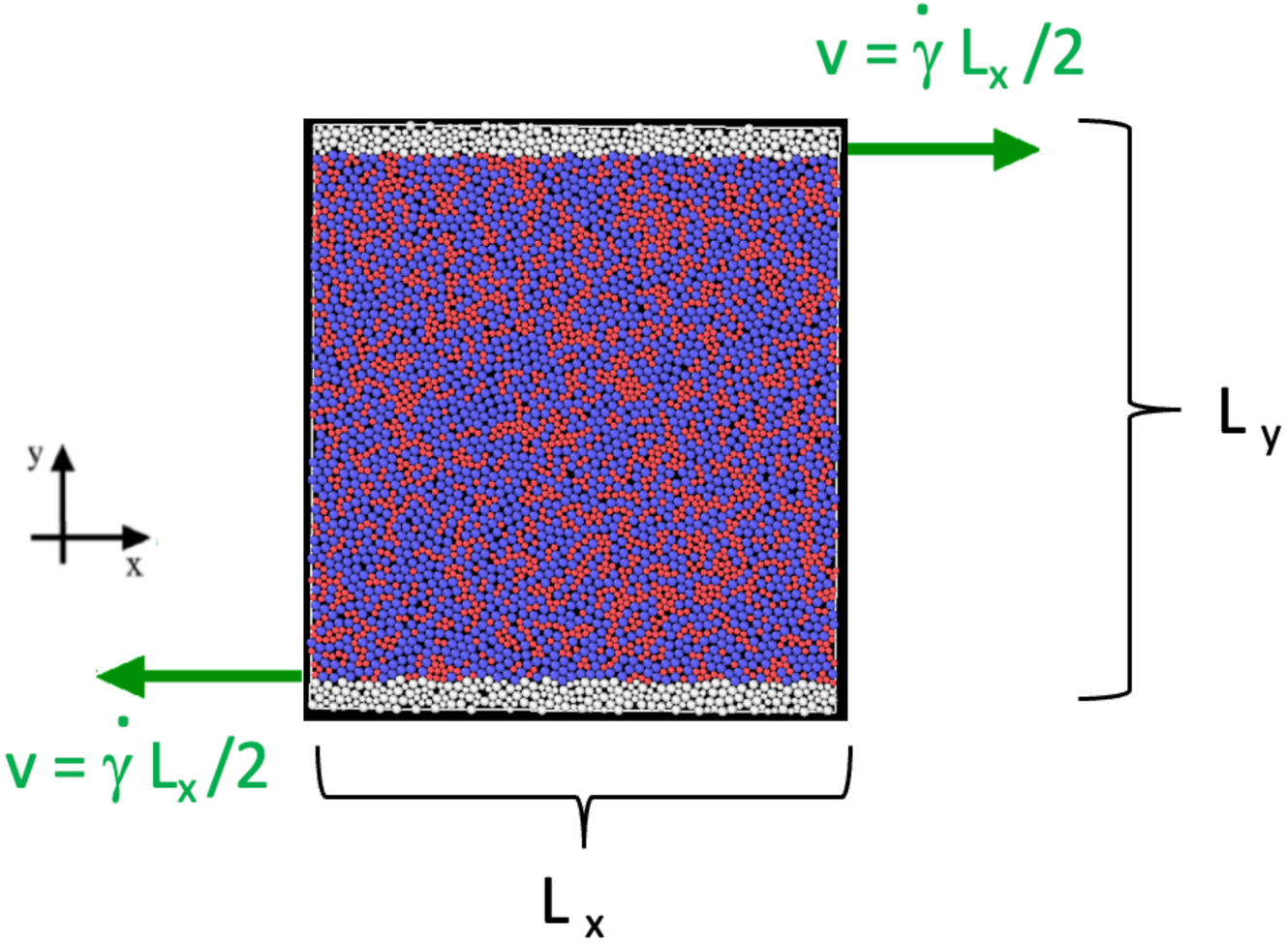
- Support vector machine (SVM) used to learn spatial patterns of softness (susceptibility to rearrangement)[1,2]
- Deep and graph neural networks shown to be well-suited for dynamics of glassy systems[3,4,5]
- Graph neural network (GNN) used to predict structure of force networks in jammed granular systems[6]

Mandal and Sollich (2022), Fig. 2: force chains in an athermal solid, with particles misidentified by the GNN in red



[1] Cubuk et al., 2015, Phys. Rev. Lett. https://doi.org/10.1103/PhysRevLett.114.108001 , [2] Ridout & Liu (2024), arXiv 2406.05868, https://doi.org/10.48550/arXiv.2406.05868, [3] Baity-Jesi et al., 2018, arXiv 1803.06969. https://doi.org/10.48550/arXiv.1803.06969, [4] Bapst et al., 2020, Nature Physics 16, 448–454. https://doi.org/10.1038/s41567-020-0842-8, [5] Wang & Zhang, 2021, Nature Communications 12, 5359. https://doi.org/10.1038/s41467-021-25490-x, [6] Mandal & Sollich (2022), Nat. Commun. 13, 4424. https://doi.org/10.1038/s41467-022-31732-3

Simulating time-evolution of system at two characteristic periods



Details about MD simulations with LAMMPS can be found in appendix